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Microprocessor Based System Design (MBSD)
6th Semester Mid-Term, Spring 2023

Max. Time: 2 hours **Max. Points: 20**

Instructions:

1. Attempt ALL questions.
2. Exam is open book and open notes.
3. Cell phones and Laptops are strictly prohibited.
4. Exam is worth **20%** of the final grade.

Q 1). **8-points**

CLO-5/PLO-3 [Cognitive Domain: Synthesis]

Design a system, where the Software in C will generate, a signal of

A. 0.25KHz with a duty cycle of 10% on P2.0 pin.

B. Whenever a user presses a button at (P3.2), the signal toggles to 0.5KHz with a duty cycle of 50%.

C. Again, pressing the same button will generate a signal of 1KHz with a duty cycle of 75%. A third time button press will result in the generation of case A and so on.

- Draw the schematic diagram showing clearly the button circuit and oscilloscope.
- Draw the timing diagram with cursors clearly showing the time period with appropriate units.
- Assuming oscillator clock of 24MHz is used.
- Use **timer** interrupt. Use External interrupts.
- Polling is not allowed.

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Q 2). 8-points

CLO-3/PLO-2 [Cognitive Domain: Application]

Fill the table by **solving** the values of variables in the program.

```
unsigned char x = 0x12, y=0, z = 0, R = 7 ;
void main(void)
 {

  while(y<5){
  x = 0x80 - y;
  x^= x;
  x += 0x7F;
  x   = x>>y;
  R += (R << 1);
  z += (x+y+R);

  y++;
  }
}
```

	X	R	Z
Y = 0	7F	15	94
Y = 1	3F	3F	13
Y = 2	1F	BD	F1
Y = 3	0F	37	3A
Y = 4	07	A5	29

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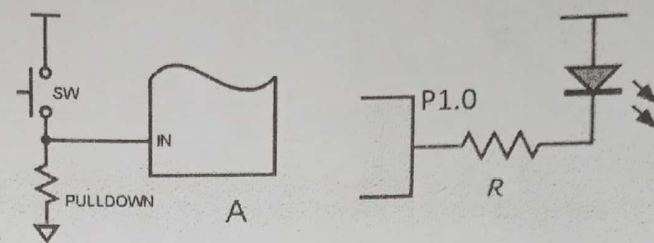
Q 3). **4-points**

CLO-2/PLO-1 [Cognitive Domain: Comprehension]

Translate the following tasks into C code.

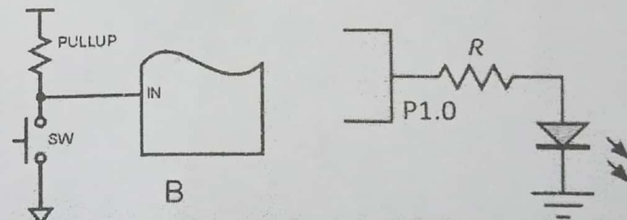
i) If we have an active-high button (A) at **P2.5** pin and an ***active-low*** *LED at P1.0* as shown below,



P2.5 = 0X04; led = 1;
Scan the button using polling
While (P2.5 == 1 *)*
{
P1.0 = 0 *// Button pressed, TURN ON the* ***active-low*** *LED at P1.0*
}

---

ii) If we have an active-low button (B) at **P2.5** and an **active-high** LED at P1.0 as shown below,



P2.5 = 0x74; led=0;
Scan the button, using polling
While (P2.5==0 *)*
{
P1.0 = 1 *// Button pressed, TURN ON the* ***active-high*** *LED at P1.0*
}